Harmonizing Synthetic and Real Remote Sensing Imagery for Object Detection via Multiscale Adversarial Training (MSAT) Framework

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***Abstract*—Recent advances in generative modeling have enabled synthetic data generation for remote sensing (RS); however, object detectors trained on synthetic–real domain gaps that limit generalization. To address this challenge, we propose multiscale adversarial training (MSAT), a unified framework that combines an SinGAN-style multiscale generator for harmonizing synthetic objects with real satellite imagery and a detector framework that performs multiscale object detection while providing realism-aware feedback to progressively improve synthetic image quality. Experiments on NWPU VHR-10, DOTA-v2.0, AID, and PatternNet demonstrate that MSAT consistently outperforms YOLOv10m, RTMDet, DINO, Swin-T + Faster R-CNN, and DETR under identical evaluation settings. MSAT achieves relative mAP@0.5 improvements of up to +12.9% on real datasets and +17.9% under hybrid training. Across settings, MSAT exhibits large relative mAP@0.5 improvements as compared to Benchmark baseline models when transitioning from real to hybrid data, from +6.50% to +12.37%, demonstrating strong robustness under domain shift. Ablation studies confirm the contribution of each proposed component, and synthetic image quality evaluation demonstrates that MSAT reduces Fréchet inception distance (FID) and learned perceptual image patch similarity (LPIPS) scores across all four hybrid datasets compared with the vanilla SinGAN baseline, indicating improved alignment with real image distributions and enhanced perceptual fidelity. These results demonstrate that realism-aware harmonization and detection-guided adversarial learning effectively reduce synthetic–real domain discrepancy and improve RS object detection. Code is publicly available at https://github.com/Faryalaurooj/Multi-Scale-Adversarial-Training-MSAT-framework**

***Index Terms*—Adversarial training, class imbalance, classification accuracy, domain adaptability, generator-discriminator framework, multiscale attention (MSA), object detection, remote sensing (RS).**

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I. Introduction

OBJECT detection in remote sensing (RS) imagery is critical for applications such as environmental monitoring, urban planning, disaster management, and defense surveillance [1], [2]. However, limited annotated data, significant variations in object scale, cluttered backgrounds, and domain shifts across sensors and imaging conditions continue to hinder robust model generalization [3], [4]. Traditional handcrafted-feature approaches have largely been replaced by deep-learning frameworks based on CNNs [5], including R-CNN [6], single shot multibox detector (SSD) [7], YOLO variants [8], [9], [10], RTMDet [11], and transformer-based detection frameworks such as DETR [12], RT-DETR [13], DINO [14], and detectors built upon Swin Transformer backbones [15], which provide strong detection accuracy and efficient feature representation. Nevertheless, these methods remain highly dependent on large-scale annotated datasets and often exhibit degraded performance under complex cross-domain RS scenarios. To address annotation scarcity, recent studies have explored synthetic data generation using generative adversarial networks (GANs) [16], style transfer [17], [18], SinGAN-based generation [19], adversarial domain adaptation [20], and diffusion-based text-to-image models such as DALL-E 2 [21], Imagen [22], and latent diffusion models [23]. Specialized RS-oriented generative frameworks, including MCGGAN [24] and spectrotemporal fusion models [25], further improve feature diversity and class balance across heterogeneous sensing conditions. In parallel, multiscale architectures such as feature pyramid networks (FPNs) [26], attention-guided GANs [27], [28], and segmentation frameworks including DeepLabv3+, SegNet, and U-Net variants [29], [30], [31], [32] have demonstrated the importance of hierarchical contextual learning for object detection.

Furthermore, many domain adaptation approaches focus primarily on feature-level alignment while neglecting structural consistency between synthetic and real imagery distributions and multiscale feature interactions [33], [34], [35], [36]. As a result, synthetic samples often introduce localization uncertainty in scene-centric datasets. Existing methods also provide limited analysis of how synthetic augmentation influences classification stability, class imbalance robustness, and cross-domain generalization under heterogeneous sensing conditions. In addition, several recent generative approaches

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rely on computationally intensive diffusion pipelines that are difficult to optimize for large-scale RS applications and real-time deployment scenarios [37], [38]. Another important gap lies in the limited integration between generative modeling and detection refinement. Most prior works treat synthetic generation and object detection as independent stages without jointly optimizing contextual realism, multiscale representation learning, and adversarial feature adaptation within a unified framework [39], [40], [41], [42], [43]. Consequently, existing methods often struggle to maintain stable performance across diverse RS datasets, particularly when transitioning from object-centric benchmarks to scene-centric environments [39], [40], [41].

Motivated by these limitations, we propose the multiscale adversarial training (MSAT) framework, which harmonizes synthetic and real RS imagery through an SinGAN-based multiscale image harmonization generator and a detection-guided realism evaluation framework. By jointly optimizing image harmonization and object detection within a unified hybrid learning pipeline, MSAT reduces the synthetic–real domain gap while improving detection robustness across heterogeneous RS datasets. We evaluate the proposed framework on multiple RGB RS benchmark datasets and across diverse detector families under both real and hybrid training settings, demonstrating consistent improvements in detection accuracy, robustness, and cross-domain generalization. Our main contributions are summarized as follows.

1) We propose MSAT, a unified hybrid learning framework for RS object detection that harmonizes synthetic objects in real RS imagery. MSAT integrates an adapted SinGAN-based multiscale image harmonization generator with a detector, incorporating multiscale feature extraction, object detection, and detection-guided realism evaluation, enabling realistic imagery synthesis while preserving scene-level contextual consistency and robust object detection under limited real-world annotations.
2) We introduce an MSAT strategy for image harmonization that progressively refines synthetic object appearance using learned background texture statistics. In addition to multiscale adversarial supervision, the generator is iteratively refined using detection-aware feedback from the detector framework, further reducing the synthetic–real domain gap. The proposed generator consistently improves the perceptual quality of synthesized samples over the baseline SinGAN model, achieving lower Fréchet inception distance (FID) and learned perceptual image patch similarity (LPIPS) scores that indicate improved structural fidelity and visual realism.
3) We show that MSAT consistently outperforms state-of-the-art detectors, including YOLOv10m, RTMDet, DINO, Swin-T + Faster R-CNN and DETR, across four RGB benchmark datasets NWPU VHR-10, DOTA-v2.0, AID, and PatternNet for detection accuracy in RS images. The greatest improvements are observed under domain-shifted hybrid scenarios (+17.9% relative improvement in mAP with Hybrid-3 as compared to YOLOv10m and +12.2% as compared to DETR), confirming the effectiveness of multiscale adversarial refinement and contextual learning in improving cross-domain generalization.
4) We demonstrate that MSAT-generated data significantly improves detection performance under hybrid training across diverse benchmarks. All baseline detectors benefit from synthetic augmentation, with relative gains up to +9.5% mAP for CNN-based models and +10.31% for transformer-based models. MSAT achieves the most consistent and substantial improvements, reaching up to a relative gain of +12.37% mAP@0.5 under hybrid settings.
5) We observe that hybrid training improves performance on structured object detection datasets (NWPU VHR-10 and DOTA-v2.0), while scene-centric datasets (AID and PatternNet) introduce negative transfer for CNN-based detectors. In contrast, MSAT maintains stable gains across all settings, demonstrating strong robustness to synthetic–real domain shift and noisy pseudolabel supervision. The code is released at https://github.com/Faryalaurooj/Multi-Scale-Adversarial-Training-MSAT-framework

This study demonstrates that MSAT improves performance at two levels: 1) it generates high-fidelity synthetic data with better perceptual quality than vanilla SinGAN and 2) it translates this improved data quality into consistent detection gains across benchmarking baseline models. The improvements are most pronounced under hybrid training settings, confirming that better synthetic realism directly enhances cross-domain generalization and robustness in RS object detection. The remainder of this article is organized as follows. Section II details the proposed framework, Section III describes the proposed methodology, including training and evaluation strategy, Section IV presents results and discussion, Section V describes limitations of this study, Section VI outlines future directions, and Section VII concludes the study.

II. Proposed Framework

MSAT combines an SinGAN-based multiscale image harmonization generator framework $G(N)$ with a detector framework $D(N)$ to improve both synthetic image quality and downstream detection performance. An overview of the workflow is shown in Fig. 1.

A. Generator: Synthetic-to-Real Image Generation

The generator operates in two sequential steps: 1) multiscale background texture learning and 2) synthetic object harmonization. Internally, these two stages are implemented through a multiscale coarse-to-fine generation pipeline. As illustrated in Fig. 1, the generator progressively constructs a background representation using an input image pyramid, multiscale noise injection, residual refinement, and upsampling operations. During background texture learning, these components learn hierarchical texture statistics from real satellite imagery. The same learned multiscale generator is subsequently reused during synthetic object harmonization to blend inserted object patches with the learned background.

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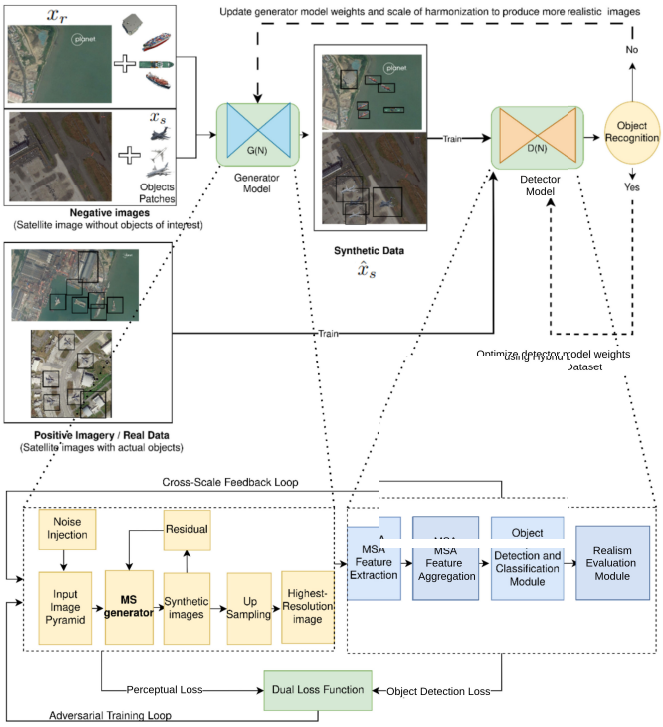


Fig. 1. Overview of the proposed MSAT framework. The generator framework $G(N)$, trained using multiscale adapted SinGAN generators G_n and discriminators D_n, harmonizes synthetic object patches with real satellite backgrounds to generate realism-enhanced images. The detector framework $D(N)$, consisting of the MSA feature extraction, MSA feature aggregation, object detection and classification, and realism evaluation and detection-aware feedback modules, performs object detection, realism evaluation, and feedback generation. Refined detections are converted into a reliability weight ω, which adaptively weights the detection loss and provides realism-aware feedback for generator refinement. Both frameworks are jointly optimized through the proposed dual-loss learning strategy.

1) Multiscale Background Texture Learning: Let x_r be a real satellite background image sampled from the data distribution p_{data}, extracted from Google Earth Pro [44], covering diverse geographic regions and acquisition conditions, including airports, urban areas, and mixed land-cover scenes. During the first stage, the adapted SinGAN-inspired multiscale framework is trained on an image pyramid $\{x_i\}_{i=0}^{n}$ obtained by progressively downsampling x_r. Internally, each pyramid level performs scale-specific noise injection, multiscale generation, residual refinement, and upsampling using multiscale generators $\{G_i\}_{i=0}^{n}$ and discriminators $\{D_i\}_{i=0}^{n}$, where $i = 0$ denotes the finest (highest-resolution) pyramid level, and $i = n$ denotes the coarsest (lowest-resolution) pyramid level (Fig. 2). At each pyramid level i, the generator G_i receives a scale-specific noise map Z_i together with the upsampled output $(x_{i+1}^{*})^{\uparrow}$ from the next coarser scale, except at the coarsest level ($i = n$), where G_n receives only the noise input Z_n and produces a synthesized image x_i^{*}. The corresponding discriminator D_i enforces patch-level realism by distinguishing generated patches from patches sampled from the real image x_i. These discriminators operate exclusively within the multiscale SinGAN training process and should not be confused with the detector framework $D(N)$ used later in MSAT. While each D_i provides adversarial supervision for image synthesis at pyramid level i, the detector framework

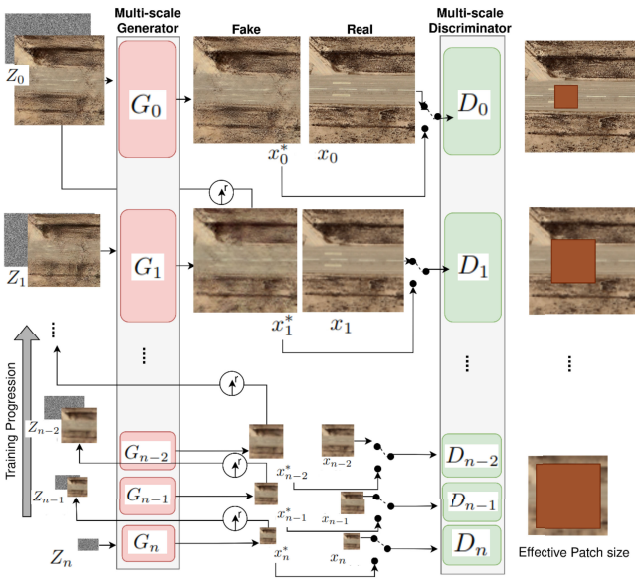


Fig. 2. Multiscale SinGAN-based pipeline. Generators $\{G_i\}_{i=0}^{n}$ and discriminators $\{D_i\}_{i=0}^{n}$ are trained progressively across image scales. At each scale i, generator G_i receives noise Z_i and the upsampled output from the previous scale $(x_{i+1}^{*})^{\uparrow}$ to generate x_i^{*}, while discriminator D_i enforces patch-level realism. The learned multiscale background model is subsequently used for synthetic object harmonization to produce the realism-enhanced image $\hat{x}_s$.

$D(N)$ evaluates the final harmonized image for realism assessment and detection-aware feedback. To preserve structural fidelity during progressive generation, a reconstruction loss based on root-mean-square error (RMSE) is employed as shown in the following equation:

$$L_{\text{RMSE}}^{(i)} = \sqrt{\frac{1}{M}\sum_{j=1}^{M}\left(x_i(j) - x_i^{*}(j)\right)^2} \tag{1}$$

where x_i and x_i^{*} denote the real and generated images at pyramid level i, respectively, and M is the total number of pixels. This loss encourages structural consistency between generated and real imagery while reducing reconstruction artifacts across scales. Training proceeds progressively from coarse to fine scales using adversarial and reconstruction objectives. The final synthesized image is produced at the highest resolution and serves as the learned background model for subsequent object harmonization.

2) Synthetic Object Harmonization: In the second step, a synthetic object patch x_s is naively inserted into satellite image x_r and harmonized using the previously trained multiscale background model. The generator transfers learned texture and illumination characteristics while preserving the structural identity of the inserted object, producing a realism-enhanced image $\hat{x}_s$ as shown in the following equation:

$$\hat{x}_s = G(x_s, Z) \tag{2}$$

where $Z = \{Z_i\}_{i=0}^{n}$ denotes the multiscale noise inputs across the pyramid levels. The harmonization process is performed in a coarse-to-fine manner, consistent with the background training pipeline. During adversarial training, the realism evaluation and feedback module within the detector framework provides object-aware feedback that updates the generator

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parameters and progressively refines the harmonization injection scale, enabling more realistic blending of synthetic objects with the surrounding background. After harmonization, the highest-resolution generated image $\hat{x}_s$ is forwarded to the detector framework $D(N)$.

B. Detector: Detection and Realism Evaluation Framework

The detector model $D(N)$ serves as the downstream detection and realism evaluation framework of MSAT. Given a harmonized image $\hat{x}_s$, the detector performs object localization and classification while simultaneously evaluating contextual consistency and visual realism through a dedicated feedback mechanism. The feedback signal is utilized to guide generator refinement during training. The detector $D(N)$ consists of four components: 1) multiscale attention (MSA) feature extraction module; 2) MSA feature aggregation module; 3) object detection and classification module; and 4) realism evaluation and detection-aware feedback module.

1) MSA Feature Extraction Module: The proposed MSA Feature Extraction module leverages hierarchical representations from a pretrained YOLOv8m CSP-based object detection backbone to capture both low-level spatial details and high-level semantic context. Given an input image, intermediate feature maps are extracted at multiple depths, producing a set of multiscale representations $\{F_k\}_{k=1}^{K}$ with varying spatial resolutions. To ensure structural consistency between synthetic and corresponding reference samples, a feature-level self-regularization mechanism is applied in the embedding space. Specifically, features derived from the backbone are used to enforce similarity between generated and source synthetic images, encouraging the preservation of semantic content and structural layout during generation. Given multiscale feature representations $\{F_k\}_{k=1}^{K}$ extracted at different spatial resolutions, direct softmax normalization across raw feature maps is not feasible due to dimensional inconsistencies across scales. To address this, each feature map is first compressed into a compact scale descriptor using global average pooling (GAP), as shown in the following equation:

$$s_k = \mathrm{GAP}(F_k) \tag{3}$$

where s_k represents the global response of the kth feature scale. These descriptors provide a scale-wise summary of feature activations, enabling comparable representations across heterogeneous resolutions. To model relative importance across different feature scales, a softmax normalization is applied over the pooled descriptors, yielding adaptive MSA weights as shown in the following equation:

$$a_k = \frac{\exp(s_k)}{\sum_{j=1}^{K} \exp(s_j)} \tag{4}$$

where a_k denotes the normalized attention weight for the kth feature scale, s_k is the corresponding scale descriptor from (3), K is the total number of feature scales, and j is the summation index. The softmax normalization ensures that $\sum_{k=1}^{K} a_k = 1$, allowing adaptive recalibration of multiscale feature maps as shown in the following equation:

$$\hat{F}_k = a_k \cdot F_k \tag{5}$$

where $\hat{F}_k$ denotes the attention-refined feature map at the kth scale. This adaptive weighting mechanism ensures that more informative scales contribute more strongly to the final representation, while less relevant scales are suppressed, enhancing scale-aware detection capability in complex RS scenes.

2) MSA Feature Aggregation Module: Let I be the input RS image, and let $\hat{F}_k(I)$ be the attention-enhanced feature map extracted at the kth scale after feature alignment and refinement. To aggregate multiscale contextual information, the refined features from all scales are fused through weighted summation as shown in the following equation:

$$F_{\mathrm{MSA}}(I) = \sum_{k=1}^{K} \hat{F}_k(I) \tag{6}$$

where K denotes the total number of feature scales, and $F_{\mathrm{MSA}}(I)$ denotes the aggregated multiscale feature representation. Feature maps are first aligned through interpolation and channel projection before aggregation. The resulting representation enables the detector framework $D(N)$ to capture complementary multiresolution features, improving robustness to scale variation, small objects, and background clutter. The aggregated features are then forwarded to the detection and classification stage for object localization and prediction.

3) Object Detection and Classification Module: Using the aggregated feature representation $F_{\mathrm{MSA}}(I)$, a cascading detection strategy is employed to progressively refine predictions from coarse to fine representations. Let $S_{\mathrm{coarse}} \in \mathbb{R}^{H\times W\times C}$ be the coarse detection score map produced by the detection head, where H and W represent the spatial dimensions of the feature map, and C denotes the number of object classes. The refined score map S_{fine} is computed as shown in the following equation:

$$S_{\mathrm{fine}} = \alpha S_{\mathrm{coarse}} + \beta \mathcal{F}(S_{\mathrm{coarse}}) \tag{7}$$

where $\mathcal{F}$ denotes a lightweight convolutional refinement head that learns residual corrections to the coarse detection score map, thereby improving localization accuracy while preserving the original feature dimensions. The learnable scalar coefficients α and β balance the contribution of the original detection scores and the refinement term during training. This residual refinement strategy enables adaptive correction of object confidence estimates and improves localization robustness in dense and cluttered RS scenes.

4) Realism Evaluation and Detection-Aware Feedback Module: Following object detection and classification, the realism evaluation and detection-aware feedback module evaluates whether the harmonized samples preserve visual realism, semantic consistency, and localization accuracy within the detector framework $D(N)$. The refined detection score map S_{fine} serves as an object-level reliability signal, where high-confidence predictions indicate close alignment with the real data distribution, while low-confidence predictions suggest artifacts or structural inconsistencies. The aggregated detection confidence is converted into a scalar reliability weight ω, which regulates the contribution of detection

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supervision during the dual-loss optimization, as defined in the following equation:

$$\omega = \rho\left(S_{\text{fine}}\right) \tag{8}$$

where $\rho(\cdot)$ denotes a confidence aggregation function over the refined detection scores. The resulting reliability weight ω adaptively weights the detection loss during optimization of the detector framework, as defined in (14). The optimized detector framework subsequently provides realism-aware feedback to the generator framework for improving synthetic image realism and harmonization. This module introduces no additional loss function; instead, it generates the reliability weight ω used by the dual-loss optimization.

C. Dual Loss Formulation

The proposed MSAT framework adopts a dual-loss optimization strategy in which adversarial learning is guided by two complementary objectives: 1) object-level detection supervision and 2) perceptual consistency preservation. The detector framework $D(N)$ is jointly optimized to distinguish real satellite images from harmonized synthetic images while simultaneously providing object detection, realism evaluation, and detection-aware feedback. The adversarial objective is defined in the following equation:

$$\mathcal{L}_{adv}^{D} = \mathbb{E}_{x_r \sim p_{\text{data}}}\left[\log D\left(x_r\right)\right] + \mathbb{E}_{\hat{x}_s}\left[\log\left(1 - D\left(\hat{x}_s\right)\right)\right] \tag{9}$$

where $D(\cdot)$ denotes the realism evaluation score produced by the detector framework for an input image. The term $x_r \sim p_{\text{data}}$ represents a real satellite image sampled from the real-image distribution p_{data}, while $\hat{x}_s$ denotes a harmonized synthetic image generated by the generator framework $G(N)$. The expectation operator $\mathbb{E}[\cdot]$ computes the average loss over the corresponding training samples. The first term, $\log D(x_r)$, encourages the detector framework to assign high realism confidence to real satellite images, whereas the second term, $\log(1 - D(\hat{x}_s))$, encourages it to assign low realism confidence to harmonized synthetic images that deviate from the real data distribution. The resulting realism evaluation, together with the detection-aware feedback generated by the detector framework, provides adversarial supervision for refining the generator framework. The detector framework jointly performs object detection, realism evaluation, and detection-aware feedback generation. Its object detection and classification module produces refined detection predictions S_{fine}. The corresponding detection loss is defined in the following equation:

$$\mathcal{L}_{det} = \mathcal{L}_{cls} + \mathcal{L}_{bbox} \tag{10}$$

where $\mathcal{L}_{cls}$ and $\mathcal{L}_{bbox}$ denote the classification and bounding-box regression losses, respectively. Following (8), the refined detection map S_{fine} is aggregated into a scalar reliability weight ω, which serves as a feedback signal that adaptively regulates the contribution of detection supervision during adversarial training. To preserve structural and semantic consistency between harmonized and real imagery, the generator framework is additionally constrained by a perceptual consistency loss as shown in the following equation:

$$\mathcal{L}_{perc} = \phi\left(x_r, \hat{x}_s\right) \tag{11}$$

where $\phi(\cdot)$ denotes an L1/L2 feature-space distance computed using a pretrained feature extractor. The generator objective combines adversarial and perceptual supervision as shown in the following equation:

$$\mathcal{L}_G = \mathcal{L}_{adv}^{G} + \lambda \mathcal{L}_{perc} \tag{12}$$

where

$$\mathcal{L}_{adv}^{G} = -\mathbb{E}_{\hat{x}_s}\left[\log D\left(\hat{x}_s\right)\right] \tag{13}$$

where λ controls the contribution of perceptual consistency. $\mathcal{L}_{adv}^{G}$ denotes the adversarial objective used to optimize the generator framework $G(N)$, while $D(\hat{x}_s)$ denotes the realism score assigned by the detector framework to the harmonized synthetic image. Minimizing this objective encourages the generator to produce harmonized images that are visually realistic, contextually consistent, and capable of yielding reliable object detection and high realism confidence within the detector framework. The detector framework is optimized using a combination of adversarial discrimination and reliability-weighted detection supervision as defined in the following equation:

$$\mathcal{L}_D = \mathcal{L}_{adv}^{D} + \omega \mathcal{L}_{det}. \tag{14}$$

The overall optimization problem is defined as shown in the following equations:

$$G^* = \arg\min_{G}\left(\mathcal{L}_{adv}^{G} + \lambda \mathcal{L}_{perc}\right) \tag{15a}$$

$$D^* = \arg\min_{D}\left(\mathcal{L}_{adv}^{D} + \omega \mathcal{L}_{det}\right) \tag{15b}$$

where G^* and D^* denote the optimized generator framework and detector framework obtained after convergence, respectively. In this formulation, $\mathcal{L}_{perc}$ encourages structural consistency between harmonized and real imagery, while $\omega\mathcal{L}_{det}$ provides object-aware feedback derived from refined detection confidence. Their joint optimization establishes the proposed dual-loss adversarial learning mechanism illustrated in Fig. 1, enabling the generator framework and detector framework to be optimized collaboratively for generating realistic harmonized RS imagery and improving downstream object detection performance under synthetic-to-real domain shift.

III. Proposed Methodology

The proposed framework is trained to jointly optimize synthetic image generation and object detection under synthetic–real domain shift. In the first stage, the generator framework $G(N)$ learns hierarchical background texture and scene priors using an adapted SinGAN-inspired multiscale architecture trained on real satellite imagery. This enables the model to capture scale-consistent structural and texture statistics of RS scenes. In the second stage, synthetic object patches are inserted into real satellite backgrounds, and $G(N)$ performs multiscale contextual harmonization to generate realism-enhanced synthetic images $\hat{x}_s$ while preserving both object integrity and background consistency. In the third stage, the detector framework $D(N)$ is trained using a hybrid dataset consisting of real satellite images x_r and harmonized synthetic images $\hat{x}_s$. The detector framework incorporates the

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TABLE I
DATASET SPLIT SPECIFICATIONS AND IMAGE COUNTS WITH 50:50 REAL-TO-SYNTHETIC BALANCE

Dataset	Total Images	Train (%)	Val (%)	Test (%)	Train (Count)	Val (Count)	Test (Count)
NWPU VHR-10	800	70.00%	15.00%	15.00%	560	120	120
DOTA-v2.0	11,268	16.24%	5.26%	78.50%	1,830	593	8,845
AID	10,000	50.00%	20.00%	30.00%	5,000	2,000	3,000
PatternNet	30,400	50.00%	20.00%	30.00%	15,200	6,080	9,120
Hybrid-1 (NWPU)	1,600	70.00%	15.00%	15.00%	1,120	240	240
Hybrid-2 (DOTA)	22,536	16.24%	5.26%	78.50%	3,660	1,186	17,690
Hybrid-3 (AID)	20,000	50.00%	20.00%	30.00%	10,000	4,000	6,000
Hybrid-4 (PatternNet)	60,800	50.00%	20.00%	30.00%	30,400	12,160	18,240

proposed MSA feature extraction module, MSA feature aggregation module, object detection and classification module, and realism evaluation and detection-aware feedback module to simultaneously perform object detection, evaluate synthesized image realism, and generate detection-aware feedback for adversarial training. Finally, the optimized MSAT framework is evaluated on both real and hybrid test datasets to assess its robustness and cross-domain generalization capability.

A. Benchmark Datasets for RGB-Based Analysis

We evaluate MSAT on four RGB-based RS benchmarks: **NWPU VHR-10** [45], **DOTA-v2.0** [46], **AID** [47], and **PatternNet** [48]. These datasets comprise high-resolution aerial imagery (0.3–1.0 m/pixel) with diverse scene complexity and acquisition conditions. NWPU VHR-10 and DOTA-v2.0 provide object-level annotations (e.g., airplanes, ships, vehicles, bridges, and storage tanks), while AID and PatternNet are scene-level datasets containing transportation-relevant environments such as airports, ports, parking lots, and industrial regions and lack object-level labels; therefore, pseudo bounding boxes are generated using sliding-window proposals refined by class activation maps (CAMs)-based localization. These annotations are fixed across experiments for fair comparison and enable evaluation under weak supervision and scene-level variability. The study targets transportation-related object categories including vehicles, ships, airplanes, transport aircraft, cars, and trucks. To ensure cross-dataset consistency, semantically similar labels are unified into a shared taxonomy: car and small vehicle are mapped to vehicle, while truck and large vehicle are grouped as heavy vehicle; aircraft-related labels are aligned under airplane/transport aircraft, enabling consistent evaluation across heterogeneous benchmarks without reannotation of original datasets. To assess robustness under synthetic–real domain shift, four hybrid datasets are constructed using SinGAN-based object insertion. Hybrid-1 and Hybrid-2 use NWPU VHR-10 and DOTA-v2.0 scenes, respectively, while Hybrid-3 and Hybrid-4 extend AID and PatternNet, respectively, annotations generated under weak supervision constraints. All hybrid datasets maintain a 50:50 real-to-synthetic ratio, with the number of samples in train/val/test sets shown in Table I. Fig. 3 illustrates an example of Hybrid-2, showing synthetic–real blending using the proposed multiscale synthesis strategy.

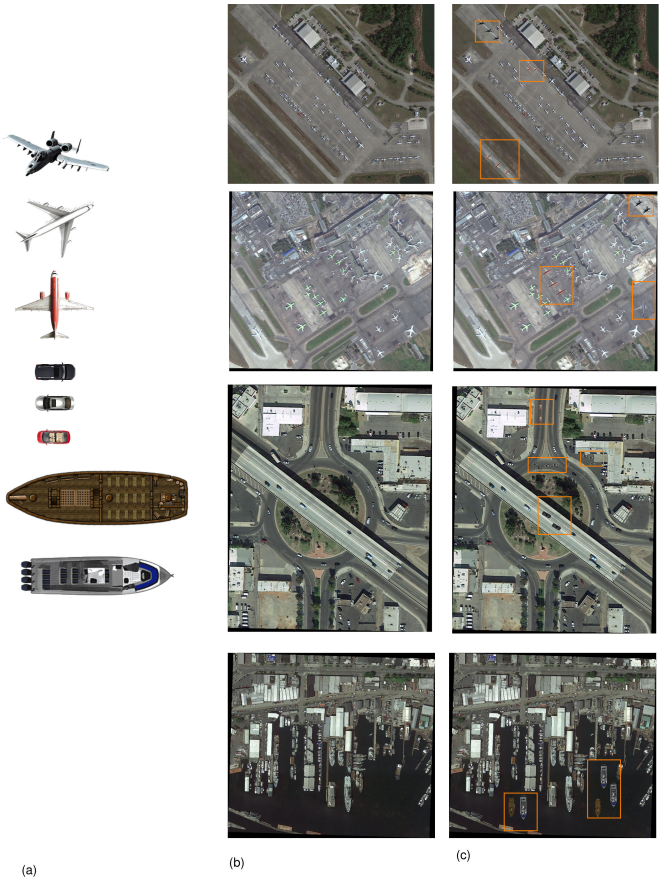


Fig. 3. Hybrid-2: containing (a) synthetic patches, (b) real satellite imagery from DOTA-v2.0 with real objects of interest, and (c) harmonized synthetic images, which are a blend of real satellite imagery with synthetic objects harmonized in backgrounds through the proposed MSAT framework.

B. Training Protocol

We consider two training regimes: 1) real-only training using annotated data from benchmarking datasets mentioned in Section III-A; and 2) hybrid training consisting of real satellite images x_r of benchmarking datasets referred to as positive imagery with actual objects and MSAT-generated harmonized synthetic images $\hat{x}_s$ referred to as negative imagery, background-only satellite images of benchmarking datasets with harmonized synthetic objects (Fig. 1). All models are evaluated on held-out test sets for fair comparison. Official dataset splits are followed for all benchmarks as shown in Table I. The training sets of Hybrid datasets (Hybrid-1 to Hybrid-4) are constructed only from training splits of real datasets to prevent data leakage, and for the val and test sets of hybrid datasets, validation and test sets of the respective real datasets are used. Dataset statistics remain fixed across experiments for reproducibility. For each real sample,

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$k = 5$ synthetic variants are produced offline. During training, mini-batches are balanced with an approximately 50:50 real-to-synthetic ratio to prevent distribution dominance. Synthetic data generation is split into an offline generation phase and an online sampling strategy. Synthetic samples are generated offline using an adapted SinGAN-style generator trained on real object crops. For each real sample, $k = 5$ synthetic variants are produced. Online, data leakage and distribution dominance are prevented during training by dynamically balancing each mini-batch to maintain an approximate 50:50 real-to-synthetic ratio, drawing randomly from the pregenerated offline pool.

C. MSAT Implementation

The generator framework $G(N)$ is trained using an eight-scale SinGAN pyramid with 2000 iterations per scale, a learning rate of 2×10^{-4}, and a noise amplitude of 0.1. During multiscale background texture learning, the corresponding SinGAN discriminators $\{D_i\}_{i=0}^{n}$ provide patch-level adversarial supervision at each pyramid scale. The detector framework $D(N)$ is trained on the hybrid dataset using the proposed realism evaluation and detection-aware feedback module, enabling implicit synthetic–real domain alignment without requiring explicit domain adaptation modules. During adversarial optimization, realism-aware feedback is propagated to the generator framework to update its parameters and refine the harmonization injection scale (1–5), improving the tradeoff between background texture transfer and object boundary preservation. The MSA module uses $K = 3$, corresponding to feature maps extracted at spatial resolutions $\{1/4, 1/8, 1/16\}$. Spatial alignment across scales is achieved through interpolation and channel projection before feature aggregation. The detector framework is trained with a batch size of 16, and the generator objective uses a perceptual consistency weight of $\lambda = 0.3$. MSAT is implemented in PyTorch and trained on an NVIDIA RTX 3090 GPU with 24 GB VRAM. The detector $D(N)$ is optimized using the AdamW optimizer with weight decay of 5×10^{-4} and cosine learning rate scheduling. The multiscale generators and discriminators inside the generator framework $G(N)$ are trained alternately with a 1:1 update ratio, and the adversarial weight $\lambda = 0.3$ is fixed across all datasets to ensure consistent evaluation. We evaluate all methods on both real and hybrid datasets under cross-domain and distribution-shift settings. For reproducibility, the SinGAN-based generator used for synthetic data generation is trained once using a fixed random seed (42), and no dataset-specific hyperparameter tuning is applied beyond class-wise training separation. The resulting hybrid datasets are kept fixed throughout all experiments. The harmonization injection scale is treated as an adaptive hyperparameter and is refined during adversarial training using the detector framework's realism-aware feedback. MSAT was implemented using YOLOv8m as the backbone in the detector framework of MSAT. The selection of YOLOv8m was made prior to the release and widespread adoption of YOLOv10. To ensure a fair assessment of performance, YOLOv10m was included as an independent state-of-the-art baseline in all comparative experiments to evaluate the effectiveness of the proposed MSAT training strategy rather than to introduce a new detector architecture.

D. Evaluation Protocol

We benchmark MSAT against five representative detectors spanning CNN-based and transformer-based paradigms: YOLOv10m, RTMDet, DINO, Swin Transformer, and DETR. CNN-based models serve as efficient one-stage baselines, while transformer-based methods capture global context through attention mechanisms. The Swin Transformer baseline is implemented using a Swin-T backbone with a Faster R-CNN detection head. All models are trained and evaluated under identical input resolution, augmentation, learning rate schedules, and optimization settings to ensure a fair comparison, highlighting MSAT's robustness across heterogeneous architectures. For inference speed evaluation, all models were exported to TensorRT and executed using FP16 precision on a single NVIDIA RTX 3090 GPU. Frames per second (FPS) was measured with a batch size equal to one using identical input resolution (640×640) and optimization settings for all methods. To eliminate initialization overhead, 100 warm-up iterations were performed before timing, and the reported FPS values were obtained by averaging over 1000 consecutive inference iterations.

E. Performance Metrics

Evaluation is conducted using metrics to assess detection accuracy, robustness, and efficiency. Detection performance is measured using mean average precision at an IoU threshold of 0.5 (mAP@0.5) along with $F1$-score to capture localization–classification tradeoffs. To evaluate robustness under class imbalance, we report G-mean and balanced accuracy, computed from class-wise true/false positives and negatives using a fixed IoU threshold and averaged across categories. Computational efficiency is measured using FPS, while cross-domain robustness is evaluated on both real and hybrid datasets. Discriminator training and evaluation experiments were repeated over three independent runs using different random seeds. Mean performance is reported, while the standard deviation of mAP@0.5 is additionally provided to quantify run-to-run variability and robustness.

IV. Results and Discussion

A. Performance Across Real and Hybrid Datasets

1) Real Datasets: MSAT shows consistent but dataset-dependent improvements in NWPU VHR-10, DOTA-v2.0, AID, and PatternNet compared to benchmark models. The gains are more pronounced on challenging detection-oriented datasets such as DOTA-v2.0 (+12.8% relative gain), where complex object distributions and dense layouts amplify the benefits of multiscale and semantic-aware modeling. In contrast, improvements in scene-derived datasets such as AID and PatternNet are comparatively smaller (almost + 7.3% relative gain) due to higher label saturation and reduced object-level ambiguity (Tables II and III and Figs. 4 and 5). This confirms that MSAT is particularly effective in scenarios with strong spatial complexity and fine-grained object interactions rather than uniformly across all domains. AID and PatternNet are scene-classification datasets and do not provide native object-level bounding-box annotations. Therefore, object detection

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TABLE II

COMPREHENSIVE PERFORMANCE COMPARISON ACROSS REAL AND HYBRID DATASETS. HYBRID-1 TO HYBRID-4 ARE CONSTRUCTED AS SYNTHETIC + NWPU VHR-10, SYNTHETIC + DOTA-V2.0, SYNTHETIC + AID, AND SYNTHETIC + PATTERNNET, RESPECTIVELY. SINCE AID AND PATTERNNET DO NOT PROVIDE OBJECT-LEVEL DETECTION ANNOTATIONS, WEAK OBJECT-LEVEL PSEUDO-LABELS ARE GENERATED USING CAMS FOR EVALUATION PURPOSES. REPORTED DETECTION METRICS ON HYBRID-3 AND HYBRID-4 ARE APPROXIMATE INDICATORS OF OBJECT LOCALIZATION PERFORMANCE. MAP@0.5, $F1$-SCORE, G-MEAN, AND BALANCED ACCURACY ARE REPORTED IN PERCENTAGES. G-MEAN AND BALANCED ACCURACY ARE COMPUTED VIA CLASS-WISE AVERAGING BY CONVERTING OBJECT-LEVEL PREDICTIONS INTO BINARY OUTCOMES (TP/TN/FP/FN) UNDER A FIXED IOU THRESHOLD. FPS IS MEASURED USING TENSORRT FP16 INFERENCE ON AN NVIDIA RTX 3090 GPU WITH BATCH SIZE = 1 AFTER 100 WARM-UP ITERATIONS AND AVERAGED OVER 1000 INFERENCE ITERATIONS

Model	Dataset	mAP@0.5 (%) ±Std	F1 (%)	G-mean (%)	Bal. Acc. (%)	FPS
YOLOv10m	NWPU	61.8 ± 0.48	63.4	59.0	61.0	56
	DOTA-v2.0	60.9 ± 0.47	62.4	58.0	60.2	
	AID	66.5 ± 0.45	67.5	63.0	66.0	
	PatternNet	67.2 ± 0.48	68.3	64.0	66.8	
	Hybrid-1	65.5 ± 0.44	66.5	64.0	65.0	
	Hybrid-2	66.7 ± 0.46	67.5	65.0	66.2	
	Hybrid-3	66.0 ± 0.53	66.5	64.0	65.5	
	Hybrid-4	66.4 ± 0.50	67.0	65.0	65.9	
RTMDet	NWPU	62.3 ± 0.50	64.0	60.0	61.5	21
	DOTA-v2.0	61.2 ± 0.53	63.0	59.0	60.4	
	AID	66.8 ± 0.46	68.3	64.0	66.3	
	PatternNet	67.5 ± 0.44	69.3	65.0	67.1	
	Hybrid-1	65.8 ± 0.48	66.5	64.0	65.2	
	Hybrid-2	67.0 ± 0.45	67.5	65.0	66.5	
	Hybrid-3	66.4 ± 0.47	66.5	64.0	65.9	
	Hybrid-4	67.2 ± 0.48	67.0	65.0	66.7	
DINO	NWPU	64.9 ± 0.42	66.3	62.0	64.5	25
	DOTA-v2.0	64.0 ± 0.44	65.3	61.0	63.7	
	AID	68.2 ± 0.38	69.3	65.0	67.8	
	PatternNet	69.0 ± 0.37	70.3	66.0	68.6	
	Hybrid-1	69.5 ± 0.35	71.3	69.0	69.2	
	Hybrid-2	70.6 ± 0.34	72.3	70.0	70.3	
	Hybrid-3	70.9 ± 0.33	73.3	71.0	70.6	
	Hybrid-4	70.4 ± 0.35	72.3	70.0	70.1	
Swin-T + Faster R-CNN	NWPU	65.4 ± 0.40	67.3	63.0	65.0	35
	DOTA-v2.0	64.3 ± 0.42	66.3	62.0	63.9	
	AID	68.8 ± 0.36	70.3	66.0	68.4	
	PatternNet	69.6 ± 0.35	71.3	67.0	69.2	
	Hybrid-1	69.9 ± 0.34	72.3	70.0	69.5	
	Hybrid-2	70.8 ± 0.33	73.3	71.0	70.4	
	Hybrid-3	70.5 ± 0.34	72.3	70.0	70.1	
	Hybrid-4	71.0 ± 0.32	73.3	71.0	70.6	
DETR	NWPU	63.9 ± 0.48	65.3	61.0	63.4	18
	DOTA-v2.0	62.8 ± 0.50	64.3	60.0	62.2	
	AID	67.1 ± 0.44	68.3	64.0	66.7	
	PatternNet	67.9 ± 0.43	69.3	65.0	67.5	
	Hybrid-1	68.0 ± 0.42	70.3	68.0	67.6	
	Hybrid-2	69.2 ± 0.40	71.3	69.0	68.8	
	Hybrid-3	68.8 ± 0.41	70.3	68.0	68.4	
	Hybrid-4	69.4 ± 0.40	71.3	69.0	69.0	
MSAT (Ours)	NWPU	69.8 ± 0.34	72.3	68.0	69.5	40
	DOTA-v2.0	68.7 ± 0.35	71.3	67.0	68.3	
	AID	71.5 ± 0.30	74.3	70.0	71.2	
	PatternNet	72.3 ± 0.29	75.3	71.0	72.0	
	Hybrid-1	75.8 ± 0.27	78.3	76.0	75.5	
	Hybrid-2	77.2 ± 0.25	80.3	78.0	77.0	
	Hybrid-3	77.8 ± 0.24	**81.3**	**79.0**	**77.6**	
	Hybrid-4	77.0 ± 0.26	79.3	77.0	76.8	

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TABLE III

ABSOLUTE MAP@0.5 (%) IMPROVEMENTS OF MSAT OVER BASELINE DETECTORS ACROSS REAL AND HYBRID DATASETS (MSAT–BASELINE). VALUES OUTSIDE PARENTHESIS REPRESENT ABSOLUTE DIFFERENCES IN MAP@0.5 (%) AND VALUES INSIDE PARENTHESIS REPRESENT RELATIVE GAINS IN MAP@0.5% COMPUTED UNDER IDENTICAL TRAINING AND EVALUATION PROTOCOLS

MSAT gain over model	NWPU	DOTA-v2.0	AID	PatternNet	Hybrid-1	Hybrid-2	Hybrid-3	Hybrid-4
YOLOv10m	**+8.0 (12.9%)**	**+7.8 (12.8%)**	+5.0 (7.5%)	+5.1 (7.6%)	**+10.3 (15.7%)**	**+10.5 (15.7%)**	**+11.8 (17.9%)**	**+10.6 (16.0%)**
RTMDet	**+7.5 (12.0%)**	**+7.5 (12.3%)**	+4.7 (7.0%)	+4.8 (7.1%)	**+10.0 (15.2%)**	**+10.2 (15.5%)**	**+11.4 (17.2%)**	**+10.0 (14.9%)**
DINO	+4.9 (7.6%)	+4.7 (7.3%)	+3.3 (4.8%)	+3.3 (4.8%)	+6.3 (9.1%)	+6.6 (9.4%)	+6.9 (9.7%)	+7.4 (10.5%)
Swin-T + Faster R-CNN	+4.4 (6.7%)	+4.4 (6.8%)	+2.7 (3.9%)	+2.7 (3.9%)	+5.9 (8.4%)	+6.4 (9.1%)	+6.5 (9.3%)	+6.0 (8.5%)
DETR	+5.9 (9.2%)	+5.9 (9.4%)	+4.4 (6.5%)	+4.4 (6.5%)	+7.8 (11.5%)	+8.0 (11.6%)	+8.4 (12.2%)	+7.6 (11.0%)

TABLE IV

DATASET-WISE PERFORMANCE IMPROVEMENTS FROM REAL TO HYBRID TRAINING (HYBRID MINUS (−) REAL). VALUES ARE REPORTED AS ABSOLUTE DIFFERENCES IN MAP@0.5 (%), WHILE RELATIVE IMPROVEMENT IN MAP@0.5 (%) IS SHOWN IN PARENTHESES. POSITIVE VALUES INDICATE IMPROVEMENT/GAIN, AND NEGATIVE VALUES INDICATE PERFORMANCE DEGRADATION UNDER HYBRID TRAINING

Model	(Hybrid-1) - NWPU	(Hybrid-2) - DOTA-v2.0	(Hybrid-3) - AID	(Hybrid-4) - PatternNet
YOLOv10m	+3.7 (+5.99%)	+5.8 (+9.52%)	-0.5 (-0.75%)	-0.8 (-1.19%)
RTMDet	+3.5 (+5.62%)	+5.8 (+9.48%)	-0.4 (-0.60%)	-0.3 (-0.44%)
DINO	+4.6 (+7.09%)	+6.6 (+10.31%)	+2.7 (+3.96%)	+1.4 (+2.03%)
Swin-T + Faster R-CNN	+4.5 (+6.88%)	+6.5 (+10.11%)	+1.7 (+2.47%)	+1.4 (+2.01%)
DETR	+4.1 (+6.42%)	+6.4 (+10.19%)	+1.7 (+2.53%)	+1.5 (+2.21%)
MSAT (Ours)	**+6.0 (+8.60%)**	**+8.5 (+12.37%)**	**+6.3 (+8.81%)**	**+4.7 (+6.50%)**

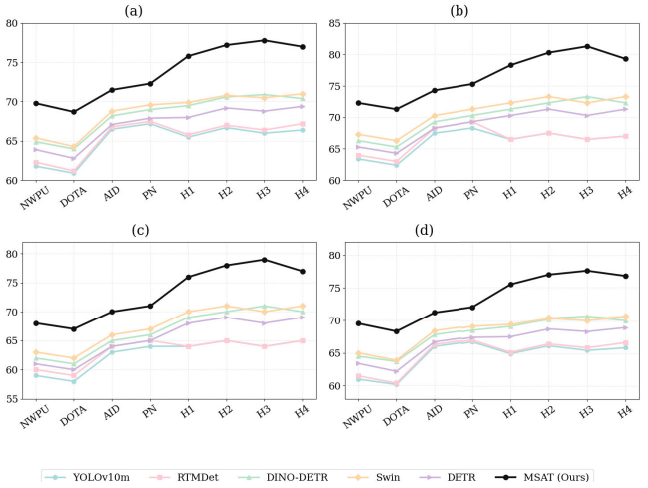


Fig. 4. Comprehensive performance comparison of MSAT across baseline models across real and hybrid datasets. Hybrid-1 (H1) to Hybrid-4 (H4) denote Synthetic + NWPU VHR-10, Synthetic + DOTA-v2.0, and Synthetic + AID, Synthetic + PatternNet (with weak pseudoannotations), respectively. Evaluation metrics include (a) mAP@0.5 (%), (b) $F1$-score (%), (c) G-mean (%), and (d) balanced accuracy (%).

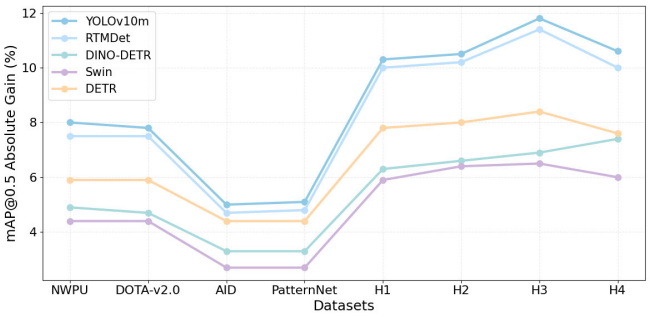


Fig. 5. Absolute mAP@0.5 improvements achieved by MSAT over baseline detectors across real and hybrid datasets [Hybrid-1 (H1) to Hybrid-4 (H4)].

metrics reported on Hybrid-3 and Hybrid-4 in Table II are computed using CAM-derived pseudolabels. While these results provide useful evidence of relative model behavior under weak supervision, they may not fully reflect true object detection performance and should be interpreted with this limitation. All detection experiments were repeated three times using different random seeds. Table II reports the mean and standard deviation of mAP@0.5 across runs. The relatively small standard deviations (typically below 0.5 percentage points) indicate stable training behavior and low run-to-run variability. Owing to the substantial computational cost, formal statistical significance testing and confidence interval analysis were not conducted. Performance gains achieved by MSAT over the strongest baseline are consistently larger than the observed standard deviations, suggesting that the improvements are meaningful and unlikely to arise solely from random initialization effects.

2) Hybrid Datasets: All models benefit from hybrid synthetic augmentation, though the magnitude of improvement varies across architectures and datasets (Table IV and Fig. 6). Hybrid-2 (DOTA-v2.0) yields the most consistent gains, reflecting the effectiveness of object-level synthetic supervision in structured, object-centric scenes. In contrast, Hybrid-4 (PatternNet) shows smaller improvements or slight degradation for CNN-based baselines, attributed to increased noise from weak pseudoannotations and the dominance of scene-level semantics over object-level cues. Specifically, CNN-based detectors (YOLOv10m, RTMDet) exhibit mild performance drops on Hybrid-3 and Hybrid-4 due to localization uncertainty introduced by synthetic insertion and weak supervision; however, transformer-based models remain more stable across

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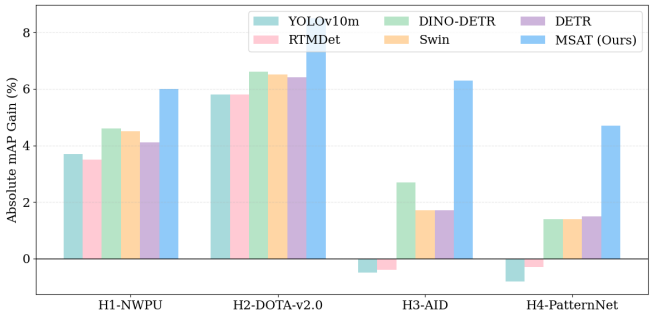


Fig. 6. Hybrid training performance improvement (absolute mAP improvements) (hybrid–real) across four benchmark datasets (H1-NWPU, H2-DOTA-v2.0, H3-AID, and H4-PatternNet). The proposed framework consistently achieves higher gains compared to baseline detectors, demonstrating its effectiveness under hybrid training settings.

datasets. In contrast, MSAT achieves consistent improvements across all settings (+8.60% on Hybrid-1, +12.37% on Hybrid-2, +8.81% on Hybrid-3, and +6.50% on Hybrid-4), demonstrating improved robustness to synthetic–real domain shift in both object-centric and scene-centric scenarios.

3) Detection Accuracy and Classification Stability: The proposed framework achieves consistent and robust performance across both real and hybrid datasets, maintaining strong precision–recall balance under all evaluation settings (Table III and Fig. 5). On real datasets, it attains mAP@0.5 scores of 69.8 (NWPU VHR-10), 68.7 (DOTA-v2.0), 71.5 (AID), and 72.3 (PatternNet), which further improve under hybrid training to 75.8 (Hybrid-1), 77.2 (Hybrid-2), 77.8 (Hybrid-3), and 77.0 (Hybrid-4). Similar trends are observed for $F1$-score, G-mean, and balanced accuracy, confirming improved class-wise stability under imbalance-sensitive evaluation. Compared to CNN-based detectors (YOLOv10m, RTMDet), which show more variability and weaker hybrid gains, and transformer-based models (DETR, Swin-T + Faster R-CNN, DINO), which remain competitive but consistently lower, MSAT demonstrates superior generalization across all datasets. The largest improvements are observed in Hybrid-2 and Hybrid-3, highlighting the benefit of synthetic augmentation under domain shift and weak supervision. Overall, MSAT consistently outperforms all baselines across mAP@0.5, $F1$-score, G-mean, and balanced accuracy, particularly in structurally complex datasets such as DOTA-v2.0 and PatternNet.

4) Efficiency and Robustness Trade-Off: A clear tradeoff is observed between computational efficiency and detection accuracy across models. YOLOv10m achieves the highest inference speed (56 FPS) but consistently underperforms in mAP@0.5 and $F1$-score across both real and hybrid settings. Transformer-based detectors and MSAT improve accuracy at the cost of moderate computational overhead, while remaining within real-time constraints. The proposed framework operates at 40 FPS due to multiscale adversarial refinement and semantic-guided modulation, yet achieves the best overall performance across all datasets. It consistently outperforms competing baselines on both real and hybrid benchmarks, with particularly large gains on Hybrid-2 (77.2%) and Hybrid-3 (77.8%), where domain shift and weak supervision are more pronounced. As runtime is independent of dataset identity under fixed input resolution and batch size, FPS is reported as a single averaged value per model.

TABLE V

QUALITY ASSESSMENT OF SYNTHETIC SAMPLES USING FID AND LPIPS ACROSS FOUR HYBRID DATASETS. EACH HYBRID SETTING CORRESPONDS TO A SPECIFIC BENCHMARK DATASET. RESULTS ARE REPORTED USING 5000 GENERATED SAMPLES PER SETTING, EVALUATED AGAINST THE CORRESPONDING REAL IMAGE DISTRIBUTIONS. LOWER VALUES INDICATE BETTER PERCEPTUAL QUALITY AND CLOSER ALIGNMENT WITH REAL DATA DISTRIBUTIONS. COMPARISON IS PROVIDED BETWEEN VANILLA SINGAN AND THE PROPOSED MSAT GENERATOR

Hybrid Setting	Method	FID ↓	LPIPS ↓
Hybrid-1 (NWPU VHR-10)	SinGAN	21.4	0.168
	MSAT (Ours)	17.6	0.142
Hybrid-2 (DOTA-v2.0)	SinGAN	19.8	0.156
	MSAT (Ours)	16.2	0.131
Hybrid-3 (AID)	SinGAN	23.1	0.174
	MSAT (Ours)	19.5	0.149
Hybrid-4 (PatternNet)	SinGAN	20.6	0.161
	MSAT (Ours)	16.9	0.137

B. Quality Assessment of Synthetic Samples

One of the primary components of the proposed framework is the generator framework, which performs multiscale background texture learning and synthetic object harmonization to produce realism-enhanced RS images. To quantitatively evaluate the quality of the harmonized synthetic samples generated by the MSAT generator, we employ two widely used perceptual image quality metrics: FID and LPIPS. FID measures the distributional similarity between generated and real images, as shown in the following equation:

$$\mathrm{FID} = \|\mu_r - \mu_s\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_s - 2\left(\Sigma_r\Sigma_s\right)^{1/2}\right) \tag{16}$$

where μ_r and μ_s are mean features, and Σ_r and Σ_s covariances of real and synthetic images; lower FID indicates closer alignment. LPIPS measures perceptual similarity as shown in the following equation:

$$\mathrm{LPIPS}(x, y) = \sum_l w_l \, \|f_l(x) - f_l(y)\|_2^2 \tag{17}$$

where f_l denotes the feature representation extracted from layer l, and w_l represents the corresponding layer weight. Lower LPIPS values indicate higher perceptual similarity and better structural fidelity between synthesized and real images. To evaluate the proposed generator framework, 5000 harmonized synthetic images are generated by MSAT for each dataset and compared with the corresponding real image distribution (NWPU VHR-10, DOTA-v2.0, AID, and PatternNet) using FID and LPIPS. The same evaluation protocol is applied to synthetic images generated by the original SinGAN, allowing a fair comparison of image realism and perceptual quality between the two frameworks. As summarized in Table V, the proposed MSAT framework consistently achieves lower FID and LPIPS scores than the baseline SinGAN across all benchmark datasets, demonstrating improved

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TABLE VI

ABLATION STUDY OF THE PROPOSED FRAMEWORK ON THE HYBRID-4 DATASET. COMPONENTS ARE INCREMENTALLY ADDED ACCORDING TO THE ARCHITECTURE DESCRIBED IN SECTION II. THE FINAL CONFIGURATION, INCLUDING THE REALISM EVALUATION AND DETECTION-AWARE FEEDBACK MODULE, REPRESENTS THE COMPLETE MSAT FRAMEWORK

Configuration	mAP@0.5 (%)	F1 Score (%)	Balanced Acc. (%)	FPS
YOLOv8m (Baseline)	55.2	56.0	54.8	47
+ SinGAN-based Generator	60.3	61.4	60.0	47
+ MSA Feature Extraction Module	65.8	67.5	65.4	45
+ MSA Feature Aggregation Module	69.4	71.3	69.0	43
+ Object Detection and Classification Module	74.8	77.0	74.5	40
+ Realism Evaluation and Detection-Aware Feedback Module (ω)	77.0	79.3	76.8	40

alignment with real image distributions and enhanced perceptual fidelity. On average, MSAT reduces FID by approximately 15%–18% and LPIPS by 10%–14%, indicating more realistic texture synthesis and improved structural consistency. These improvements are primarily attributed to the integration of the generator with the proposed detector framework, where the detector provides realism-aware adversarial feedback through its realism evaluation and feedback modules during training. This additional supervisory signal guides the generator toward producing harmonized synthetic images that are not only visually realistic but also more suitable for downstream object detection. These improvements are further supported by the ablation study, where removing the proposed MSAT components results in degraded synthetic image quality and reduced downstream object detection performance.

C. Ablation Study

We evaluate the individual and combined contributions of the individual modules and the full MSAT framework. Table VI reports an incremental analysis on the Hybrid-4 dataset.

1) *YOLOv8m (Baseline):* Baseline detector without SinGAN-based generator.
2) *+SinGAN-Based Generator:* Introduces realism-enhanced synthetic sample generation through multiscale background learning and object harmonization.
3) *+MSA Feature Extraction Module:* Applies MSA to extract scale-aware and context-aware feature representations.
4) *+MSA Feature Aggregation Module:* Aggregates attention-refined features from multiple scales into a unified representation.
5) *+Object Detection and Classification Module:* Performs coarse-to-fine object localization and classification using aggregated multiscale features.
6) *+Realism Evaluation and Detection-Aware Feedback Module:* Uses refined detection confidence to generate reliability-guided feedback that adaptively weights detection supervision during adversarial training. MSAT framework is complete with this configuration.

The ablation results presented in Table VI demonstrate the progressive contribution of each major component of the proposed framework. Starting from the YOLOv8m baseline, the detector achieves 55.2% mAP@0.5, 56.0% $F1$-score, and 54.8% balanced accuracy. Incorporating the SinGAN-based generation module improves performance to 60.3% mAP@0.5 and 61.4% $F1$-score, indicating that realism-enhanced synthetic samples increase data diversity and partially mitigate the synthetic–real domain gap. The addition of the MSA feature extraction module further increases performance to 65.8% mAP@0.5 and 67.5% $F1$-score by enabling scale-aware feature learning and improving representation quality for small and densely distributed objects. Introducing the MSA feature aggregation module yields an additional improvement to 69.4% mAP@0.5 and 69.0% balanced accuracy, demonstrating the benefit of integrating complementary contextual information across multiple feature resolutions. Adding the object detection and classification module results in a substantial increase to 74.8% mAP@0.5 and 77.0% $F1$-score. This improvement highlights the effectiveness of the proposed coarse-to-fine detection refinement strategy in enhancing localization accuracy and classification reliability. Finally, the realism evaluation and detection-aware feedback module introduces reliability-guided supervision through the adaptive weighting factor ω, increasing performance to 77.0% mAP@0.5, 79.3% $F1$-score, and 76.8% balanced accuracy. By leveraging refined detection confidence as a feedback signal, this module improves alignment between synthesized and real imagery and strengthens the adversarial training process. This final configuration corresponds to the complete MSAT framework. Overall, the complete MSAT framework achieves a gain of +21.8 percentage points in mAP@0.5 over the YOLOv8m baseline. SinGAN-based generator operates only during dataset generation and does not affect inference speed. The observed reduction in FPS is primarily attributed to the addition of multiscale feature extraction, feature aggregation, and detection refinement modules. The complete framework maintains real-time capability, achieving 40 FPS while delivering gains in detection accuracy.

V. LIMITATIONS OF EXISTING DETECTORS AND MSAT

Existing detectors, including YOLOv10m, RTMDet, DINO, Swin-T + Faster R-CNN, and DETR, exhibit limited robustness when trained with synthetic data due to their sensitivity to synthetic–real domain shifts and the absence of mechanisms that explicitly guide the generation of realism-enhanced training samples. Consequently, they often achieve inconsistent performance improvements under hybrid training. MSAT

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addresses these limitations by integrating multiscale image harmonization and a detector-guided adversarial training strategy. Despite these improvements, performance may still degrade under extreme domain shifts, such as unseen sensor modalities, rare object configurations, or highly heterogeneous environmental conditions. Nevertheless, MSAT consistently outperforms all evaluated baseline detectors while maintaining a favorable balance between detection accuracy and computational efficiency for RS applications.

VI. Future Work and Societal Impact

Future work will extend MSAT to multimodal data, incorporating thermal, IR, SAR, and textual metadata for robust, context-aware detection via cross-modal fusion and vision-language models, supporting better generalization and zero-shot recognition. For edge and UAV deployment, we will optimize inference with quantization and pruning. MSAT reduces dependence on large annotated datasets, enabling scalable use in disaster response, environmental monitoring, and infrastructure mapping. Generating realistic synthetic samples lowers annotation costs and accelerates adaptation to new regions or sensors, while responsible use ensures avoidance of misuse or inadvertent bias. MSAT framework can potentially be integrated with newer detector families, including YOLOv10 and others, which we consider an interesting direction for future work.

VII. Conclusion

This article presented MSAT, a multiscale adversarial framework for RS object detection under synthetic-to-real domain shift. The proposed method integrates adapted SinGAN-based augmentation and a detector framework for cross-domain realism evaluation. Experimental results on four benchmark datasets demonstrate that MSAT consistently improves both synthetic image quality and object detection performance over state-of-the-art CNN- and transformer-based detectors. Despite a moderate computational overhead, MSAT maintains real-time inference at approximately 40 FPS, offering a favorable accuracy–efficiency tradeoff. Future work will focus on lightweight implementations, multimodal RS, improved cross-sensor generalization, and extending the proposed training strategy to newer detector architectures.

Declaration

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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